

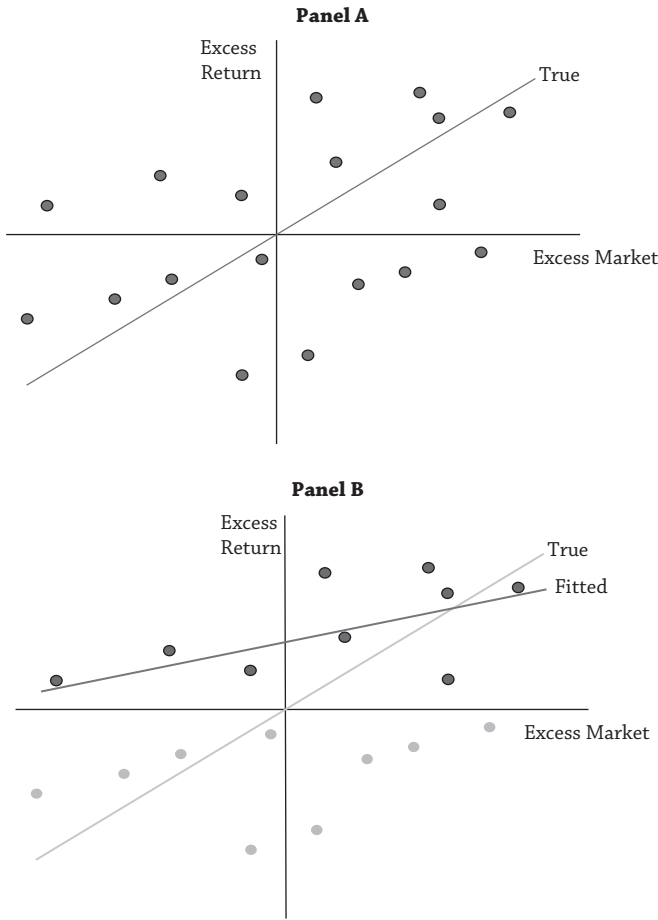


Figure 13.4

models (see chapters 6 and 7), rather than just using the market portfolio as the sole risk factor.

The effect of selection bias can be enormous. Cochrane (2005) estimates an alpha for venture capital log returns of over 90% not taking into account selection bias, which reduces to -7% correcting for the bias. Korteweg and Sorensen (2005) estimate that expected returns for the same asset class are reduced downwards by 2% to 5% per month (arithmetic returns) taking into account selection bias. The effect of selection bias in real estate is smaller, perhaps because the underlying volatility of real estate returns is lower than private equity. Fisher et al. (2003) implement selection bias corrections for real estate. They estimate that average real estate returns reduce from 1.7% to 0.3% and standard deviation estimates increase by a factor of 1.5. The small means of real estate returns are due to their sample period of 1984 to 2001, which includes the real estate downturn